

WREN

Wear with intention.

Product Requirements Document Feature Specification & Design System

Version 2.0 — AI-Powered Personal Wardrobe App
Platform: iOS & Android (React Native + Expo)
Status: MVP Development Phase
Scope: Full Product Definition

Prepared for: Engineering, Design & Academic Evaluation

1. Product Vision & Positioning

1.1 The Name: WREN

The app is named WREN — a deliberate brand decision with strategic depth. The wren is a bird renowned for being small but extraordinarily intelligent, complex, and capable — a perfect metaphor for an AI that quietly understands your entire wardrobe and surfaces exactly the right outfit with confidence.

The name was chosen over alternatives (Drape, FITT, ClosetClear) because it satisfies five critical brand criteria:

- Memorability — one syllable, instantly ownable, zero confusion in App Store search
- Fashion-forward — sits naturally alongside Aesop, Everlane, COS as a brand name
- Emotional resonance — evokes intelligence and precision without clinical coldness
- Strategic versatility — works at every scale from university project to global platform
- Narrative richness — the metaphor of a small but remarkably capable creature maps directly to the product's core value proposition

Tagline: "Wear with intention." — Communicates sustainable mindfulness, purposeful dressing, and the deliberate intelligence that WREN brings to every outfit decision.

1.2 One-Liner & North Star

WREN is the AI-powered personal stylist that lives in your pocket — turning the wardrobe you already own into a curated, intelligent, daily style experience.

North Star Metric: Daily Active Usage Rate — the percentage of users who open WREN on any given morning. This metric captures the app's core retention hypothesis: WREN must become part of the daily getting-dressed ritual, not an occasional reference tool.

Secondary Metric: Outfit Acceptance Rate — percentage of AI recommendations that users mark as "Worn Today" (target >35%).

1.3 Target User Personas

Persona 1 — Kofi, 28, Creative Professional (The Fashion Enthusiast)

Attribute	Detail
Location	Accra, Ghana
Wardrobe Size	80-120 items including statement pieces, cultural wear, streetwear
Tech Savvy	High — early adopter, uses Notion, Spotify, premium apps
Pain Points	Cannot find combinations across his large wardrobe; forgets items he owns; wants creative discovery
Goals	Discover new combinations; build a recognizable personal aesthetic; track outfit history
Motivation	Identity and self-expression; standing out thoughtfully
Key WREN Features	Style Feed, Outfit DNA, Virtual Try-On, Style Intelligence cards

Persona 2 — Amara, 34, Marketing Executive (The Busy Professional)

Attribute	Detail
Location	Lagos, Nigeria
Wardrobe Size	60-80 items; mix of business and smart casual
Tech Savvy	Medium-high — productivity app user, values efficiency above all
Pain Points	Morning decision fatigue; wastes 15+ minutes choosing outfits; repeats same combinations
Goals	30-second morning outfit decision; week planned on Sunday; never repeat within the same week
Motivation	Time and mental energy reclaimed for what matters
Key WREN Features	Home "Morning Brief", Weekly Planner, Decisive Mode recommendations

Persona 3 — Esi, 24, University Student (The Conscious Minimalist)

Attribute	Detail
Location	Accra, Ghana
Wardrobe Size	40-60 items; limited budget, buying less but better
Tech Savvy	Medium — heavy TikTok/Instagram user, not a productivity-app user
Pain Points	Cannot afford wrong purchases; wants to wear everything she owns; intimidated by luxury apps

Goals	Make every item earn its place; understand wardrobe utilization; shop smarter not more
Motivation	Sustainability values + budget consciousness
Key WREN Features	Wear intelligence, Style Feed insights, Outfit potential scores, Monthly Wrapped

1.4 Competitive Landscape

App	Strengths	Weaknesses	WREN Advantage
Stylebook	Mature wardrobe organisation	No AI, manual-only, outdated UX	AI auto-tagging + smart recommendations
Smart Closet	Clean interface	Limited intelligence, no style education	Style Feed with fashion intelligence
Cladwell	Capsule focus	Rigid, prescriptive, no personal AI	Flexible + learns user taste over time
Pinterest	Visual inspiration	External content, drives purchasing	Your wardrobe only — sustainable by design
Depop/Vinted	Secondhand community	Selling focus, not styling	Styling-first with sustainability integration

1.5 Unique Value Proposition

WREN is the only wardrobe app that combines three things no competitor offers together: (1) AI that generates outfit intelligence from your actual owned items, (2) a Style Feed that makes wardrobe browsing as engaging as social media, and (3) a Virtual Try-On that creates shareable, editorial-quality flat-lay imagery from your own clothes. The result is an app that reduces fashion consumption while making users feel more stylish — solving sustainability and desire simultaneously.

2. Design System & Visual Language

2.1 Brand & Design Philosophy

WREN's visual language sits at the intersection of premium fashion editorial and intelligent technology. Every design decision is governed by five core principles:

- **Restrained Confidence** — Show one great thing, done perfectly. Not ten adequate things. Confidence is a feature. The hero outfit recommendation is singular and bold; the Style Feed surfaces one intelligence card at a time.
- **Data Tells Stories** — Numbers without narrative are ignored. Every data point in WREN is presented as a story: "Your Oxford shirt unlocks 14 outfits" not "Outfit count: 14." This transforms analytics from a dashboard into a personal narrative.
- **Dark and Light in Dialogue** — WREN uses a deliberate dark/light contrast system. The warm ivory background is the app's resting state. Dark obsidian cards signal importance, intelligence, and drama — used for hero cards, scan screen, splash, and result screens. This visual language creates rhythm and makes key moments feel cinematic.
- **Every Screen Earns a Daily Visit** — A screen that doesn't give users a reason to return tomorrow is a dead screen. Home, Feed, and Closet are each designed with daily return hooks: weather context, spotlight items, and personalized intelligence cards.
- **Curation Over Management** — WREN's language is editorial, not administrative. Users don't "manage" their wardrobe — they "curate" it. They don't "track" outfits — they "log a wear." The language throughout the app reinforces that this is a creative, personal experience.

2.2 Color Palette

Primary Brand Colors

Token	Hex	Name	Usage
--color-bg-primary	#F5F0E8	Warm Ivory	App background, default screen surface
--color-bg-pure	#FDFAF5	Cream White	Card backgrounds, modal base
--color-bg-dark	#1A1A2E	Obsidian	Hero cards, nav pill, primary text
--color-bg-splash	#0F0F14	Near Black	Splash screen, result screens only

--color-accent-primary	#7A9E7E	Sage Green	Primary CTAs, active states, camera FAB
--color-accent-secondary	#C9A99A	Dusty Rose	Secondary accents, tags, soft alerts

Semantic / Functional Colors

Token	Hex	Usage
--color-success	#4CAF7D	Scan confirmed, saved, completion states
--color-warning	#E8A838	Low confidence tags, weather indicators
--color-error	#D95F52	Scan failure, validation errors
--color-info	#5B9BD5	Tips, tooltips, on-body try-on badge

Neutral Scale

Token	Hex	Usage
--color-text-primary	#1A1A2E	All primary body text
--color-text-secondary	#6B6B6B	Secondary text, captions, descriptions
--color-text-muted	#A8A8A8	Placeholder text, inactive tabs
--color-border	#E5E5E5	Card borders, dividers, input borders
--color-selected-bg	#EEF4EE	Selected state background (sage tint)

2.3 Typography System

Font Pairing Rationale

Display & Headers: Cormorant Garamond — Elegant, editorial, fashion-forward. Used for all screen titles, hero text, outfit names, and brand moments. Creates the luxury feel without being cold.

Body & UI: Inter — Modern, highly legible at small sizes, works perfectly for mobile labels, buttons, tags, and data. Industry standard for premium product design.

Brand Moments: Cormorant Garamond Italic — Reserved for taglines, style notes, and pull quotes. The italic creates movement against upright Inter letterforms.

Type Scale

Token	Font	Size	Weight	Line Height	Usage
display-xl	Cormorant	48px	600	1.15	Splash screen, onboarding hero
display-lg	Cormorant	36px	600	1.2	Quiz completion, major reveals
h1	Cormorant	32px	600	1.3	Screen titles (My Closet, Outfits)
h2	Cormorant	26px	600	1.3	Section heroes, card headlines
h3	Cormorant	22px	500	1.4	Sub-sections, outfit names
body-lg	Inter	16px	400	1.6	Style notes, onboarding subtext
body-md	Inter	14px	400	1.55	Card descriptions, item info
label-lg	Inter	14px	600	1.2	Buttons, active tab labels
label-sm	Inter	12px	600	1.2	Badges, occasion chips
caption	Inter	11px	400	1.5	Metadata, timestamps
overline	Inter	10px	700	1.0	Section labels (ALL CAPS, 2px tracking)

2.4 Spacing & Grid System

Base unit: 4px. All spacing values are multiples of this base. Screen horizontal padding: 24px. Bottom safe area: 40px. Status bar: 56px.

Token	Value	Primary Usage
space-1	4px	Micro gaps, icon internal spacing
space-2	8px	Icon padding, chip gaps, tight rows
space-3	12px	Compact card padding
space-4	16px	Standard component padding
space-5	20px	Card inner padding
space-6	24px	Screen padding, section gaps
space-8	32px	Large section spacing
space-10	40px	Bottom safe area, hero sections
space-12	48px	Screen vertical rhythm

2.5 Border Radius System

Token	Value	Usage
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radius-sm	6px	Tags, chips, small inputs
radius-md	12px	Buttons, small cards
radius-lg	20px	Standard cards, item cards
radius-xl	24px	Feature cards, hero cards
radius-2xl	28px	Bottom sheets, modals, splash elements
radius-full	9999px	Pills, chips, avatars, toggle elements
radius-squircle	20px	Camera FAB button (squircle — not full circle)

2.6 Shadow & Elevation

Token	CSS Value	Usage
shadow-sm	0 1px 3px rgba(0,0,0,0.08)	Default cards, subtle elevation
shadow-md	0 4px 16px rgba(0,0,0,0.10)	Active cards, floating elements
shadow-lg	0 8px 32px rgba(0,0,0,0.14)	Modals, bottom sheets
shadow-brand	0 4px 20px rgba(122,158,126,0.30)	Sage green CTA buttons
shadow-brand-lg	0 4px 20px rgba(122,158,126,0.40)	Camera FAB, primary actions
shadow-dark	0 8px 40px rgba(26,26,46,0.20)	Dark hero cards
shadow-nav	0 8px 32px rgba(26,26,46,0.20)	Bottom navigator pill

2.7 Motion & Animation Principles

- **Spring Physics** — All interactive elements use spring easing: cubic-bezier(0.34, 1.56, 0.64, 1). Feels alive and physical rather than mechanical.
- **State Transitions** — 200ms for color/opacity changes. 300ms for size/position changes. Never block user interaction during animation.
- **Scroll Reveals** — Cards fade in from below as they enter viewport: opacity 0→1, translateY 16px→0, 300ms ease-out. Creates sense of content appearing for the user.
- **Haptic Feedback** — UIImpactFeedbackGenerator.light for tab switches.
UIImpactFeedbackGenerator.medium for camera tap.
UINotificationFeedbackGenerator.success for completed actions.
- **Restraint Rule** — Animations call attention to the content, never to themselves. If an animation is noticeable as an animation rather than as a natural response, it is too much.

3. Navigation Architecture

3.1 Bottom Tab Navigator — The Floating Pill

WREN uses a custom floating pill bottom navigator — a deliberate departure from the iOS/Android default tab bar. The design rationale: a constrained-width pill that floats above the screen content communicates that navigation is a tool the user picks up, not a bar that is always in the way.

Pill Tab Bar Specification

Property	Value	Rationale
Width	300px (not full screen)	Creates floating object quality vs. a bar anchored to screen
Height	64px	Comfortable tap target with visual weight
Border Radius	99px (full pill)	Pill shape — the defining visual signature of WREN navigation
Background	#1A1A2E (Obsidian)	Dark pill on warm ivory — maximum contrast, premium feel
Bottom Margin	32px from screen edge	Floats above content — does not anchor to bottom
Shadow	0 8px 32px rgba(26,26,46,0.20)	Two-layer shadow for depth and definition

Four Destination Tabs

Tab	Icon (Phosphor)	Label	Position
Home	house-simple	"Home"	Far left
Feed	squares-four	"Feed"	Second left
Closet	hanger	"Closet"	Second right
Profile	user-circle	"Profile"	Far right

Active Tab State: An inner pill (pill-within-pill design) expands around the active icon — background #F5F0E8 (ivory), icon and label color #1A1A2E (obsidian). This inversion against the dark pill creates a deeply satisfying active state. The inner pill expands from icon-only width to icon+label width with a spring animation when a tab is selected.

Center Gap: The center 72px of the pill is intentionally empty — this gap frames the Camera FAB above it and creates a natural visual anchor.

Camera FAB — The Action Button

Property	Value	Rationale
Shape	64x64px squircle (radius 20px)	Distinguishes action from destination tabs — neither circle nor square
Background	#7A9E7E (Sage Green)	Only accent color in the navigator — unmissable, draws the eye
Icon	Phosphor camera, 26px, white (regular weight)	Outline camera feels more elegant than filled at this size
Shadow	0 8px 24px rgba(122,158,126,0.45)	Green-tinted glow — button appears as a light source
Position	Centered, 10px above pill top	Floats between content and navigation worlds
Idle Animation	Scale 1.0→1.03→1.0, 2.5s loop	Breathing pulse — subconscious invitation to tap

Scroll Behavior

Navigator hides on scroll down (translateY 0→120px, opacity 1→0, 250ms ease-in) and returns on scroll up or pause (translateY 120px→0, opacity 0→1, 300ms ease-out spring). Camera FAB and pill animate as a single unit. Navigator always visible on Home screen and when scrolled to top of any screen.

4. Screen Specifications

4.1 Splash Screen — "The Opening"

Purpose: Brand imprinting. Creates emotional anticipation before a single feature is encountered. Communicates craft and intentionality in the first 2.4 seconds.

Background: #0F0F14 (Near Black) — the "before" state. The warm ivory of the app is the "after." The contrast is the brand moment.

Visual Composition

- **The Wren Mark** — A single minimal wren bird rendered as geometric line art. Stroke weight 1.5px, color #F5F0E8, fill: none. 80x80px centered. The bird is constructed from 5 geometric shapes capturing the wren's signature raised tail — the most characterful detail that makes it unmistakably a wren.
- **Ambient Glow** — A radial gradient behind the mark: center rgba(122,158,126,0.08) → edge transparent, 200x200px, 40px blur. Barely perceptible consciously, felt subconsciously as warmth.
- **Wordmark** — "WREN" in Cormorant Garamond weight 300 (intentionally light), 36px, #F5F0E8, letter-spacing 12px. Wide tracking signals luxury. Centered, 20px below bird mark.
- **Tagline** — "Wear with intention." in Cormorant Garamond Italic 400 16px, rgba(245,240,232,0.40). Present but whispered.
- **Loading Indicator** — A minimal 48px horizontal line (1.5px height) with a sage-to-dusty-rose gradient fill animating left to right. The most restrained possible loading indicator.

Animation Sequence (Total: 2.4 seconds)

Act	Timing	What Happens
Act 1: Emergence	0ms → 600ms	Bird mark draws itself via SVG stroke-dasharray animation, starting at beak tip, tracing to raised tail. Ambient glow fades in simultaneously.
Act 2: Breath	600ms → 900ms	Single heartbeat pulse: scale 1.0→1.06→1.0, spring easing. Glow pulses in sync. The bird feels alive. This 300ms is the most important in the entire animation.
Act 3: Name	900ms → 1400ms	"WREN" materializes letter-by-letter with 60ms stagger and 8px upward drift. Tagline fades in after N completes.

Act 4: Hold	1400ms → 2400ms	Perfect stillness. 1000ms hold. Loading bar is the only movement. Stillness gives users time to absorb the brand.
Exit: Dissolve	2400ms → 2700ms	Background cross-fades #0F0F14 → #F5F0E8. Bird transforms (stroke dark), wordmark retreats upward. App world emerges from the same warm space.

4.2 Onboarding Style Quiz — "Getting to Know You"

Purpose: Collect 6 key preference signals across 5 screens to solve the cold-start problem — enabling personalized recommendations from the first scan. Framed as a style quiz, not a form.

Philosophy: Each screen should feel like a page from a luxury fashion magazine. Tap-to-select cards, not dropdowns. Visual-first, text-secondary. "Skip for now" always visible.

Quiz Flow Overview

Screen	Questions	Input Type	Data Collected
Screen 0: Introduction	None — entry gate	CTA tap	Opt-in to quiz vs. skip
Screen 1: Style Identity	Q1: Personal style archetype	Multi-select cards (6 options)	styleArchetype[]
Screen 1B: Color Comfort	Q2: Color palette preference	Single-select cards (3 options)	colorPreference
Screen 2: Lifestyle	Q3: Typical week / Q4: Climate zone	Multi-select cards / single-select cards	lifestyleType[], climateZone
Screen 3: Wardrobe Categories	Q5: Clothing categories worn	Multi-select pill cloud	categoryInclusions[]
Screen 3B: App Goal	Q6: Primary reason for using WREN	Single-select cards (4 options)	primaryGoal
Screen 4: Completion	None — profile reveal	CTA tap	Triggers profile creation

The Six Style Archetypes (Question 1)

Archetype	Descriptor	Algorithm Impact
Minimalist	Clean, neutral, simple	Weight neutral color combinations, penalise busy patterns
Smart Casual	Put-together everyday	Formality range 2-3, prioritise versatile pieces

Business	Polished and professional	Formality range 3-4, suppress ultra-casual suggestions
Streetwear	Bold, urban, relaxed	Formality range 1-2, allow bold color combinations
Maximalist	Expressive, colorful, bold	Enable complementary color pairings, embrace pattern
Eclectic	Mix of everything I love	Wide formality range, diverse combination scoring

Completion Screen: Dark background (#1A1A2E) for dramatic reveal. User's style profile shown as pill chips on a frosted glass card. Blurred outfit previews suggest what awaits. Primary CTA: "Start Scanning My Wardrobe." This is the emotional peak of onboarding — the moment the app proves it was worth setting up.

4.3 Home Screen — "The Morning Brief"

Purpose: The daily re-engagement anchor. Provides a confident single outfit recommendation with contextual intelligence (weather, occasion, behavioral nudges). The reason users open WREN every morning.

Design Reference: Calm app (contextual daily moment) × Spotify (confident recommendation) × Apple Fitness (stat cards + narrative).

Screen Sections (top to bottom, scrollable)

Section	Content	Key Design Decision
Top Bar (fixed)	Greeting with user name + notification bell + avatar	Cormorant Garamond name creates warm, personal greeting
Weather Strip	Temperature + city + fabric recommendation	Full-width pill with sage-to-dusty gradient — app feels alive and aware
Occasion Filter	"Dressing for:" + 5 horizontal chips: Work, Casual, Evening, Date, Travel	Active chip = dark (#1A1A2E) — instantly re-filters hero recommendation
Hero Outfit Card	Full-bleed flat-lay + outfit name + style note + DNA signals + Wear CTA	Dark card (obsidian) on warm ivory — the dominant visual anchor of the screen
Quick Stats Row	3 cards: Items count, Outfits today, Unworn count	Unworn card has dusty rose accent — soft behavioral nudge without alarm

Behavioral Nudge	Contextual insight card: e.g., "You've worn the white Oxford 3 days running"	Sage green icon + plain language — feels like a real stylist noticed something
More Outfits	"More for today" section header + horizontal scroll of 3 mini cards	Partial 3rd card visible — suggests scroll, drives exploration

Hero Outfit Card — Critical Element

The hero card is 380px tall with an obsidian (#1A1A2E) background. The top 260px is a full-bleed editorial flat-lay photograph of the outfit items arranged on a warm off-white surface. The bottom 120px contains the outfit name (Cormorant Garamond 26px, ivory), a three-line plain-English style note explaining WHY the combination works ("The Oxford + khaki is timeless — the brown loafers add warmth and stop it looking too corporate"), and three DNA confidence signals (Color Harmony ✓, Formality ✓, Weather ✓) as frosted glass pills.

This style note is the highest-value element in the entire app. It teaches users fashion intelligence while validating the AI's recommendation — building trust through explanation rather than assertion.

4.4 Style Feed — "The Feed"

Purpose: Daily discovery and fashion education. A vertically scrolling masonry grid that blends three content types: AI-generated outfit combinations from the user's wardrobe, personalized style intelligence derived from wear data, and visually rich fashion education that always connects back to owned items.

Design Reference: Pinterest (layout rhythm) × Spotify Discover Weekly (personalized intelligence) × Monocle magazine (editorial quality).

Content Type System

Card Type	Frequency	Width	Purpose
Your Combination	40% of feed	Single column (174px)	AI-generated outfit from user's actual wardrobe items — discovery mode
Style Intelligence	20% of feed	Full width (breakout)	Dark obsidian cards — personalized insight from wear data

Fashion Principle	20% of feed	Single or full width	Educational content with direct connection to user's owned pieces
Monthly Wrapped	Once monthly	Full width (pinned)	Shareable achievement card — Spotify Wrapped for wardrobe

Style Intelligence Card Variations

Variation	Icon	Example Headline
Power Piece	star	"Your White Oxford is your most versatile item — it unlocks 14 combinations"
Color Story	palette	"Your wardrobe speaks in blues and earth tones — one terracotta piece would unlock 8 new looks"
Wear Gap	warning	"3x more tops than bottoms is limiting your outfits — two neutral bottoms would triple your options"
Utilization	chart-pie	"You've worn 68% of your wardrobe in February — top 15% of WREN users"

Feed Architecture

- Filter chips at top: "♦ For You" | "My Combinations" | "Style Tips" | "Insights"
- Masonry 2-column grid with 10px gap — variable card heights create organic editorial rhythm
- Full-width intelligence cards break the 2-column rhythm — creates visual hierarchy
- Infinite scroll with fade-in card reveals as content enters viewport
- Save cards to named collections: "Outfit Ideas," "Style Rules I Love," "For Work Week"
- Feed refreshes: daily (new combination), weekly (new intelligence card), evergreen (fashion principles)

4.5 Wardrobe Gallery — "The Studio"

Purpose: A creative workspace for exploring, discovering, and understanding the wardrobe — not a filing cabinet. Three distinct view modes serve different mental models.

Three View Modes

Mode	What It Shows	Who It Serves	Key Feature
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Items View	Variable-height grid of individual clothing items	All users — default entry	Outfit potential badge + wear heat bar on every card
Outfits View	Pre-built outfit combinations as flat-lay cards	Fashion-forward users thinking in outfits	Outfit DNA pills showing formality, color harmony, weather
Collections View	User-named curated groups of outfits	Power users and high-engagement users	Pinterest-style curation with collage thumbnails

Item Card Intelligence Layers

- **Outfit Potential Badge** — Top-right floating badge: "8 outfits" showing how many combinations this item enables. The most novel data point in any wardrobe app. Teaches versatility, creates attachment to high-scoring items, generates mild concern about low-scoring ones — all driving curation behavior without ever saying "declutter."
- **Wear Heat Bar** — A thin 3px bar under each card. Fills proportionally to wear frequency vs. the user's most-worn item. Color signals wear level: sage green (high), dusty rose (medium), empty track with dusty tint (never worn). Glancing at the gallery shows wear distribution without reading a single number.
- **Outfit DNA Color Dots** — Three 8px colored circles overlaid at the bottom of each item image showing the item's detected color palette. Creates a visual language that runs consistently across gallery, feed, and recommendations.

Daily Spotlight Card

A dark obsidian card at the top of the gallery surfaces one item the AI wants the user to rediscover: "You haven't worn your Olive Chinos in 18 days — 4 new outfit ideas →". This single card answers "why would I open the gallery today?" — the most important retention question for any secondary screen.

Smart Sorting Options

Sort Mode	Logic	Behavioral Effect
Most Versatile	Sorts by outfit potential score	Users discover their most valuable items aren't always the most expensive
Needs Attention	Items not recently suggested in recommendations	Surfaces forgotten items — drives re-engagement with existing wardrobe
Recently Added	Standard chronological sort	Quick access to newly scanned items

Seasonal Relevance	For tropical users: fabric weight rather than calendar season	Surfaces lightweight items during hot periods
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4.6 Outfit Recommendations Screen

Purpose: Serve two distinct user modes: Decisive Mode (30-second morning outfit selection) and Exploratory Mode (Sunday evening planning, creative discovery). The screen is designed so both modes coexist without conflict.

Core Design Principle: Confidence is a feature. One great recommendation is worth more than twelve adequate ones. The hero card is not "option 1 of 5" — it is THE recommendation.

Screen Sections

Section	Purpose	Mode Served
Sticky Header	Screen title + calendar icon + filter + weather context chip	Both
Weekly Planner Strip	7-day horizontal strip with outfit thumbnail slots per day	Exploratory — Sunday ritual hook
Hero "Wear This Today" Card	Single confident recommendation with flat-lay + style note + DNA signals	Decisive — primary morning use case
"Tweak This" System	3 swap rows (shirt/trousers/shoes) with alternative thumbnails from wardrobe	Exploratory — creative participation
Outfit Memory Banner	Contextual nudge about repetition patterns	Both — behavioral intelligence
Occasion Browsing Rows	"For Work This Week" + "For the Weekend" horizontal scroll sections	Exploratory — contextual discovery
Post-Wear Rating Sheet	Bottom sheet after "Wear Today" tap — 3 outcome cards	Both — algorithm feedback loop

The Weekly Planner — Retention Mechanism

A horizontal 7-day strip showing outfit thumbnails or empty slots for each day. The Sunday ritual of planning the week drives daily app opens: users who plan Monday's outfit on Sunday have a

high probability of opening the app Monday morning to confirm and log the wear. This commitment mechanic separates apps used for a month from apps used for years.

The 3-Outcome Rating System

After logging a wear event, a bottom sheet presents three reaction cards rather than stars. This provides richer algorithm data than any rating scale:

Outcome	Icon	Signals to Algorithm
Got compliments 	chat-centered-dots	Social performance — outfit worked externally; prioritise similar combinations for social contexts
Felt great all day ✓	check-circle	Personal comfort and confidence — prioritise similar fabric/formality combinations
Wear again 	arrow-counter-clockwise	Repeat intent — high repeat value; surface combination more frequently

4.7 Virtual Try-On Screen

Purpose: WREN's most visually ambitious feature and primary word-of-mouth growth driver. Every result is designed to be screenshot-worthy and shareable — creating branded user-generated content organically.

Strategic Value: Each flat-lay export carries the WREN watermark into group chats, Instagram stories, and social media — free marketing at exactly the right demographic without a marketing budget.

Two Modes

Mode	Technology	Output Quality	Primary Use Case
Flat-Lay Studio (Default)	Image composition from item photos on warm off-white surface	Consistently premium — always beautiful	Daily use, sharing, outfit planning
On-Body Try-On (Premium)	AI compositing of outfit onto user-provided photo with 3 variations	"Wow" moment — the showstopper feature	Social sharing, outfit experimentation

Flat-Lay Studio Flow

1. User selects outfit from recommendations or builds manually in the studio canvas

2. Canvas workspace: item cards are draggable, rotatable, scalable on a warm off-white surface
3. Canvas toolbar: Arrange | Swap item | Remove | AI Enhance (repositions for editorial composition)
4. "Generate Flat-Lay" button triggers AI composition — dark → ivory background transition signals the reveal
5. Result screen: dark background (#0F0F14) for cinematic reveal of the generated flat-lay
6. Outfit DNA panel slides up below the result: three dimension bars (color harmony, formality, weather) + style note
7. Actions: Save to Collection | Wear Today | Share (triggers export card generation)

On-Body Try-On Flow

8. User uploads photo or takes a new one — privacy note: processed on-device, never stored
9. Processing screen: split composition (user photo left, scanning animation right)
10. Processing fills wait time with wardrobe insights: "Did you know? Your Oxford creates 14 combinations"
11. Result: user photo with outfit composited — 3 variation thumbnails (tucked/untucked/jacket added)
12. Comparison toggle: "Try-On" vs "Original" — the reveal of the comparison is deeply satisfying
13. Same DNA panel and action buttons as flat-lay result

Shareable Export Card

Dimensions: 1080x1080px (Instagram square). Layout: flat-lay composition on warm off-white background (60% of card), outfit name in Cormorant Garamond below, item names row, WREN bird mark + wordmark at bottom — tasteful, not dominant. The result looks like a fashion brand's Instagram post, not a screenshot. Users want to share it.

5. AI Feature Deep Dives

5.1 Wardrobe Digitization — Scan Pipeline

Step	Technology	Output
Step 1: Image Capture	React Native Camera (expo-camera) with guided frame overlay	RAW image file
Step 2: Preprocessing	On-device: background removal, lighting normalisation, crop to garment bounding box	Cleaned garment image
Step 3: Classification	Google Cloud Vision API (V1) or custom MobileNet fine-tuned on DeepFashion (V2)	category, subcategory, colours (hex), pattern, material, brand
Step 4: Confidence Gating	Category <60% → disambiguation prompt. Brand <50% → "Unknown". Colour always reliable.	Validated tag set
Step 5: Tag Review	All tags shown as editable chips — user corrections stored for retraining	Confirmed item record
Step 6: Versioning	Original AI output + user-corrected version both stored	Accuracy measurement enabled

5.2 Outfit Generation Engine

Inputs

- Full wardrobe item metadata: category, colour hex, formality score, season, material
- Current weather from OpenWeatherMap API: temperature, precipitation, wind
- Occasion context: explicit user selection from occasion chips or inferred from calendar
- Recent wear history: last 14 days — avoids combination repetition
- User style preferences from onboarding quiz and feedback history
- Mood input (V2): emoji selector mapping to style parameters

Processing — Rule-Based Layer (MVP)

14. Filter eligible items: exclude dirty/in-wash, apply seasonal appropriateness, check category availability
15. Generate combinations: top × bottom × optional layer × shoes × optional accessory
16. Score each combination across 5 dimensions:
 - Colour harmony score — complementary, analogous, neutral-anchor rules
 - Formality match score — delta ≤1 on 4-point scale (1=casual, 4=formal)
 - Weather appropriateness — temperature bands, fabric weight, waterproofing
 - Diversity score — penalise items worn in last 7 days
 - Style profile alignment — weighted against onboarding quiz answers + feedback history
17. Return top 5 combinations by composite score
18. Generate plain-English rationale for top result using style note template

Processing — ML Enhancement Layer (V2)

- Fine-tuned fashion compatibility model based on Polyvore/DeepFashion dataset approach
- Collaborative filtering: "Users with similar style profiles also loved this combination"
- Visual compatibility scoring via image embeddings — items that look good together beyond colour theory

5.3 Style Learning Loop

Signal	Type	Weight	Algorithm Effect
Mark as Worn	Explicit positive	High	Increase weight of all items and rules used in that outfit
Save Outfit	Explicit positive	High	Favourite combination — surface more frequently
Got compliments (rating)	Explicit positive	High	Social context signal — prioritise for similar occasions
Felt great (rating)	Explicit positive	Medium-High	Comfort signal — prioritise similar fabric/formality
Wear again (rating)	Explicit positive	High	Repeat intent — increase future frequency
Outfit dismissed (swipe)	Explicit negative	Medium	Reduce weight of dominant combination rule
Style quiz answers	Explicit	Medium	Sets initial preference baseline

Outfit viewed >3s	Implicit positive	Low	Noted interest — slight preference increase
Outfit card skipped	Implicit negative	Low-negative	Slight preference decrease for combination type

5.4 Outfit DNA System

Outfit DNA is WREN's visual language for communicating outfit intelligence. Applied consistently across the gallery, feed, recommendations, and try-on screens, it creates a design signature that makes WREN instantly recognizable.

For Individual Items: Three colour dots showing the item's detected colour palette — rendered as 8px circles in the item's detected colours. Instantly communicates colour story at a glance.

For Outfit Combinations: Three horizontal dimension bars shown as pill indicators: Color Harmony (complementary/analogous/neutral), Formality Position (1-4 scale), Weather Suitability (temperature range). Each bar fills proportionally to score.

For Style Notes: Every outfit recommendation includes a plain-English rationale written in the voice of a knowledgeable friend: "The Oxford + khaki pairing is timeless — the brown loafers add warmth and stop it looking too corporate." This teaches fashion intelligence while validating the recommendation.

6. Complete Feature Set — MoSCoW Prioritisation

Priority	Feature	User Story Summary	Success Metric
M — Must	Wardrobe Digitization (Scan + AI Tagging)	Scan clothing items with camera; AI auto-tags category, colour, pattern, material	Scan-to-save completion >80%; AI accuracy >85%
M — Must	AI Outfit Generation Engine	Receive 5 scored outfit suggestions filtered by weather, occasion, style preferences	Outfit acceptance rate >35%; user rating >3.8/5.0
M — Must	Style Profile & Onboarding Quiz	6-question style quiz collecting archetype, colour preference, lifestyle, climate, categories, goal	Quiz completion >75%; profile accuracy self-rating >4.0
M — Must	Home Screen "Morning Brief"	Single confident daily recommendation with weather context and occasion filter	DAU rate target; morning open-rate tracking
M — Must	Wear Logging	One-tap "Worn Today" from hero card or outfit detail — feeds all analytics	Wear log rate >60% of outfit acceptances
S — Should	Style Feed (Masonry Grid)	Scrollable feed of outfit combinations, intelligence cards, and fashion education	Feed session length >3 min; card save rate >15%
S — Should	Wardrobe Gallery — Three View Modes	Browse wardrobe as items, outfits, or collections — with outfit potential + wear heat data	Gallery return rate (daily opens) >40%
S — Should	Weekly Outfit Planner	7-day drag-assign outfit planner — the Sunday ritual retention hook	Weekly plan completion rate >30% of active users
S — Should	Virtual Try-On — Flat-Lay Studio	AI-composed editorial flat-lay of any outfit from wardrobe items — shareable export	Share rate from try-on screen >20%
S — Should	Post-Wear Rating System	3-outcome rating after logging wear: Got compliments / Felt great / Wear again	Rating completion rate >50% of logged wears
S — Should	Outfit Collections	User-named saved outfit groups: "Work Week", "Going Out"	Collection creation rate >25% of users in month 1
C — Could	On-Body Virtual Try-On	AI compositing of outfit onto user-provided photo with 3 variations	Premium feature — conversion metric for paid tier
C — Could	Monthly Wrapped Card	Shareable monthly achievement card: items worn, outfits created, utilization %	Share rate >40% of monthly active users
C — Could	Outfit Memory System	Behavioral nudges based on detected repetition and wear patterns	Nudge click-through rate >25%

C — Could	Style Insights Natural Language Search	Type "something smart but not too formal for Friday" — semantic search across wardrobe	V2 feature — search usage tracking
W — Won't (MVP)	Community / Lookbook	Share outfits with community; Style Check authentic photo prompts	Post-launch social features
W — Won't (MVP)	Shopping Gap Analysis	Identify wardrobe gaps; secondhand- first purchase recommendations	V2 with affiliate integration
W — Won't (MVP)	Packing List	Trip-based outfit planning from wardrobe	V2 feature
W — Won't (MVP)	Capsule Wardrobe Builder	AI-assisted capsule creation with coverage scoring	V2 feature

7. Technical Frontend Architecture

7.1 Recommended Tech Stack

Layer	Technology	Rationale
Mobile Framework	React Native with Expo (Managed Workflow)	Shared codebase iOS + Android; JavaScript ecosystem; Expo accelerates development; existing team knowledge
State Management	Zustand (primary) + React Query (server state)	Zustand lightweight for global UI state; React Query optimal for wardrobe data caching and sync
Navigation	React Navigation v6 (Stack + Custom Bottom Tab)	Flexible enough for custom pill navigator; deep link support; well-documented
Animation	Reanimated 2+ with Rive (for splash + complex animations)	Reanimated for 60fps UI animations on native thread; Rive for splash bird draw animation with physics
Image Handling	expo-image (caching) + expo-camera + expo-media-library	Optimised for heavy image apps; offline caching; camera access
AI Vision API	Google Cloud Vision API (V1); custom MobileNet V2 (V2)	Vision API for reliable MVP; fine-tuned model for improved accuracy and African fashion recognition
Backend	Node.js + Express REST API	JavaScript alignment with frontend; rapid development; strong ecosystem
Database	MongoDB Atlas	Flexible schema ideal for evolving item metadata; geographically close to Accra via AWS regions
File Storage	AWS S3 (item photos + flat-lay exports)	Reliable, scalable, cost-effective; CDN integration for fast image delivery
Auth	JWT + Refresh tokens + expo-secure-store	Secure local token storage; stateless auth scales well
Weather	OpenWeatherMap API (free tier: 60 calls/min)	Sufficient for MVP; location-aware; well-documented

7.2 Key Technical Challenges & Solutions

Challenge	Proposed Solution
Image-heavy performance	Lazy loading with expo-image; thumbnail generation on upload (400px for grid, 800px for detail); WebP format; CDN delivery via CloudFront
AI scan accuracy for African fashion	Google Vision for base classification; manual correction data stored for fine-tuning; explicit "Unknown" fallback with user correction flow

Cold-start problem	Onboarding quiz provides baseline preferences before first item scanned; outfit engine has fallback mode for <10 items
Offline functionality	Wardrobe data cached locally with React Query; scan queued offline and uploaded when connectivity restored; key screens function without network
Flat-lay image composition	Server-side image compositing using Sharp.js; item PNG extraction with background removal API; editorial arrangement algorithm
60fps animations on mid-range devices	All animations on native thread via Reanimated; avoid JS-thread bridge for animation logic; test on Tecno/Infinix devices for West African market

8. Core Data Models

8.1 User

```
{ _id, email, name, createdAt, styleProfile: { archetypes[], colorPreference, lifestyleTypes[], climateZone, categoryInclusions[], primaryGoal }, preferences: { notificationsEnabled, weeklyPlannerReminder, feedRefreshTime }, subscription: { tier, expiresAt } }
```

8.2 ClothingItem

```
{ _id, userId, imageUrl, thumbnailUrl, category, subcategory, colours: [{ hex, name, dominance }], pattern, material, brand, formalityScore (1-4), season[], tags[], outfitPotentialScore, wearCount, lastWornAt, addedAt, aiVersion, userCorrected: Boolean, scanConfidence }
```

8.3 Outfit

```
{ _id, userId, items: [ClothingItem._id], name, styleNote, occasion, weatherSuitability: { minTemp, maxTemp }, colorHarmonyScore, formalityScore, dnaSignals: { colorHarmony, formality, weatherFit }, isAIGenerated, createdAt, savedAt, lastWornAt }
```

8.4 WearEvent

```
{ _id, userId, outfitId, wornAt, rating: { outcome, timestamp } (outcome: "compliments" | "feltGreat" | "wearAgain"), weather: { temp, condition }, occasion }
```

8.5 StyleProfile (embedded in User)

```
{ archetypes: ["minimalist" | "smartCasual" | "business" | "streetwear" | "maximalist" | "eclectic"], colorPreference: "neutral" | "balanced" | "bold", lifestyleTypes: ["office" | "casual" | "social" | "mixed"], climateZone: "tropical" | "temperate" | "cold", categoryInclusions: String[], primaryGoal: "outfits" | "declutter" | "tracking" | "all" }
```

9. Monetisation Strategy

9.1 Freemium Model

Feature	Free Tier	WREN Pro (GH¢20/month)
Wardrobe Items	Up to 30 items	Unlimited
Daily Outfit Suggestions	3 per day	Unlimited + "Surprise Me"
Style Feed	Fashion Principles cards only	Full feed incl. Your Combinations + Insights
Virtual Try-On	Not available	Flat-Lay Studio + On-Body Try-On
Weekly Planner	View only	Full drag-assign planning
Outfit Collections	2 collections max	Unlimited collections
Monthly Wrapped	Summary only	Full shareable export card
Outfit History	30 days	Full history
AI Accuracy	Standard Vision API	Enhanced fine-tuned model (V2)

9.2 Retention Hooks

- **Daily Brief notification** — 7am push: "Good morning Kofi — your outfit for today is ready." Single outfit recommendation. Deep links to Home screen.
- **Sunday Planner reminder** — "Plan your week in 5 minutes →" — drives the Sunday ritual engagement.
- **Spotlight notification** — Occasional: "You haven't worn your Burgundy Blazer in 3 weeks — here's how to style it today."
- **Monthly Wrapped release** — End of month notification: "Your February Wrapped is ready." Drives opens and sharing.
- **Wardrobe milestone** — "You've scanned 25 items! Your outfit options just doubled."

10. Launch Strategy & Roadmap

10.1 MVP — Day 1 Feature Set

- Splash screen with animated wren mark
- Onboarding style quiz (6 questions, 5 screens)
- Camera scan with Google Vision API tagging
- Tag review and edit screen
- Home screen "Morning Brief" with weather integration
- AI outfit generation engine (rule-based, 5 combinations)
- Outfit recommendations with style notes
- Basic wardrobe gallery (Items view only)
- Wear logging (one-tap from outfit card)
- Post-wear 3-outcome rating
- Custom floating pill bottom navigator
- User profile and style preferences editor

10.2 V1 — Month 2-3

- Style Feed (full masonry grid with all 4 card types)
- Wardrobe Gallery Outfits and Collections view modes
- Virtual Try-On — Flat-Lay Studio
- Weekly Outfit Planner
- Outfit Collections (named saved groups)
- Monthly Wrapped generation and sharing
- Outfit Memory behavioral nudge system
- Push notifications (daily brief + spotlight)

10.3 V2 — Month 4-6

- On-Body Virtual Try-On with variation selector
- ML-enhanced outfit engine (fine-tuned compatibility model)
- Natural language wardrobe search

- Shopping Gap Analysis (secondhand-first)
- Community Lookbook with Style Check mechanic
- Packing List for trips
- Custom CNN fine-tuned on DeepFashion + African fashion dataset
- Capsule Wardrobe Builder

10.4 Key Metrics by Stage

Stage	Metric	Target
MVP Launch	Onboarding quiz completion	>75%
MVP Launch	Scan-to-save completion rate	>80%
MVP Launch	AI tagging accuracy (category)	>85%
MVP Launch	Day 7 retention	>40%
V1	Outfit acceptance rate (worn/suggested)	>35%
V1	DAU / MAU ratio	>25%
V1	Style Feed session length	>3 minutes
V1	Virtual Try-On share rate	>20%
V2	Day 30 retention	>25%
V2	Monthly Wrapped share rate	>40%
V2	Pro tier conversion rate	>5% of MAU

11. Design Inspiration References

App / Brand	Specific Lesson Applied to WREN
Spotify Wrapped	Monthly Wrapped card — personalized data told as a story, designed for sharing. Proves data can be emotional rather than analytical.
Spotify Daily Mix	The confident single recommendation model. One curated choice presented with conviction beats a list of options that creates decision fatigue.
Airbnb	Photography standards as trust mechanism. WREN's flat-lay quality directly determines perceived AI intelligence — bad images = bad trust.
Apple Fitness	Stat cards with narrative context. Numbers become motivating when paired with personal benchmarks and progress framing.
Duolingo	Streak mechanics and daily ritual creation. The Weekly Planner creates a Sunday ritual; the Daily Brief creates a morning ritual.
Notion	"Building block" philosophy. WREN's collections, planner, and studio are flexible tools users shape to their own system.
Calm	Contextual daily moment. Every morning the app knows what day it is, what the weather is, and what the user needs. WREN's Morning Brief is this for fashion.
Pinterest	Masonry grid visual discovery. The Style Feed borrows the layout rhythm but makes it personal — your wardrobe, not the internet's.
BeReal	Authenticity mechanic. The Style Check feature (V2) — showing what you're actually wearing, unfiltered — builds genuine community.
Linear	Precision navigation. The floating pill navigator feels like a precision tool, not a default component. Every interaction is considered.
A24	Brand restraint. The splash screen takes inspiration from A24's studio card — one symbol, total confidence, no explanation needed.
Net-a-Porter	Fashion editorial as product experience. WREN's item photography guidelines, flat-lay aesthetics, and outfit presentation borrow the fashion authority of luxury retail.

12. Academic Context & Novel Contributions

12.1 Why WREN Merits First Class Honors

Criterion	WREN's Evidence
Technical Depth	Integrates 4 CS domains: Mobile Development (React Native/Expo), Computer Vision (Google Vision API + MobileNet fine-tuning), Algorithm Design (multi-factor scoring engine), Data Visualization (Style Feed, analytics)
Real-World Impact	Promotes sustainable fashion behaviour aligned with UN SDG 12 (Responsible Consumption). Measurable: wear logging data quantifies reduction in wardrobe underutilization.
Novel Contribution	Outfit DNA as a UI pattern — visual communication of colour harmony, formality, and weather suitability as a consistent design language across all screens. No existing wardrobe app implements this.
Research Potential	Virtual try-on + wear logging creates a dataset correlating AI outfit recommendations with actual wear decisions — publishable behavioural insight.
Design Innovation	Full design system (colour tokens, typography scale, component library, motion principles) documented at production quality. Custom floating pill navigator is a genuine design contribution.
Cross-Cultural Consideration	Tropical climate zone as primary use case (Accra context). African fashion recognition gap explicitly documented with mitigation strategy. User study targets $n \geq 20$ locally relevant participants.

12.2 Limitations & Honest Assessment

- Wear Logging Burden — relies on manual user input; passive tracking not feasible in MVP. Acknowledged in dissertation with Apple Watch / Android Wear integration as V2 direction.
- Vision API Cultural Bias — Google Vision trained predominantly on Western fashion. Lower accuracy expected for Kente cloth, Ankara prints, and other African textiles. Manual correction flow is the mitigation; diverse dataset is the long-term solution.
- Cold-Start Problem — addressed by onboarding quiz providing baseline preferences before any items are scanned. Documented as a known challenge with explicit solution.
- Flat-Lay Quality Dependency — outfit recommendation quality is partially dependent on item photo quality. Camera guidance UI and photography tips in onboarding mitigate this.

WREN

Wear with intention.

Product Requirements Document v2.0

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